

ECON-7102

Environmental Economics I

Course details*

Prof. Casey J. Wichman
Georgia Institute of Technology

Fall 2026

This document accompanies the posted course syllabus. The syllabus governs course policies. This document covers assignments, the class schedule, and where to find the readings. It will be updated during the semester. Changes will be announced in class.

1 Course information

Instructor: Prof. Casey J. Wichman (wichman@gatech.edu)
Course: ECON 7102, Environmental Economics I
Term: Fall 2026
Meeting time: Mon/Wed, 11:00 AM – 12:15 PM
Classroom: TBD
Office: Old Civil Engineering 310
Office hours: By appointment
Course website: <https://caseyjwichman.com/econ7102/>

2 Course assignments

Attendance and participation

Attendance is expected and will be necessary to succeed in the course. Participation in class discussions is strongly encouraged. Ask questions. If you do not understand something in class, you are almost certainly not the only one. If you never volunteer an incorrect answer, you aren't taking enough risks.

Attendance is not graded directly. Poor performance in this dimension may affect the final grade.

*Companion to the posted course syllabus. Prof. Casey Wichman: wichman@gatech.edu, School of Economics, Georgia Institute of Technology, 221 Bobby Dodd Way, Atlanta, GA 30332. **Last update:** August 5, 2026.

“Weekly” reading responses (20 points)

Students write a short response paper, roughly 250–300 words or two paragraphs, on designated readings. These readings are marked [**response due**] in the reading list. Expect eight to ten over the semester.

The response needs to go beyond summarizing the reading. Make connections to other topics in the course, to stories in the news, or to ongoing policy discussions. Reflect on how the reading updated your thinking. If you disagreed with a conclusion or a fundamental assumption, expand on that.

Focus on three questions:

1. What was one thing you enjoyed about the article?
2. What was one thing you did not understand, or would like to discuss in class?
3. What extensions or applications of the article would be interesting?

Submit via Canvas by 11:59 PM ET the day before the class in which the reading is assigned. That timing lets me read the responses before class. Late responses receive half credit.

Research paper proposal (35 points)

One goal of this course is to provide the tools necessary to conduct your own research in environmental, energy, and resource economics. Students develop a research proposal that introduces a novel and answerable research question, reviews related literature, discusses potential contributions, and begins to explore data sources and empirical methods. There is no explicit page requirement, though 10–15 pages with normal formatting is expected. The proposal should look and read like an extended introduction to an economics journal article: title, abstract, main body, background where relevant, conceptual framework, and references. Because this course focuses on economic theory applied to environmental problems, bring that conceptual thinking to your proposal even if the goal is an empirical paper.

Research takes time, with many fits and starts, so there are milestones through the semester. Start thinking about ideas as soon as possible and discuss them with your classmates and with me. Early feedback prevents rabbit holes. Topics may need revision if they are not aligned with the course or do not represent feasible projects. Ideally, proposals become third-year papers or dissertation chapters.

Milestone	Due	Points
Project idea and research sketch	October 9	5
Expanded outline and literature review	November 6	5
Presentation	Last four classes	5
Final proposal	December 11	20

Students cannot use this proposal in conjunction with a writing assignment for another course. Proposals can sit at the intersection with other fields, for example environment and health, environment and development, or environment and trade.

Midterm (25 points)

In class, Class 15.

Lead in-class discussion (10 points)

Each student leads or co-leads discussion of one paper in the topics section of the course.

Referee report on a job-market paper (10 points)

Students find a job-market paper by a current Ph.D. candidate working on an environmental, energy, or resource economics topic, and write a referee report. The goal is to see the end product of the Ph.D. on a topic of interest and to practice giving critical but constructive feedback.

I will provide guidance during the semester on where to search for job-market papers and on the structure of a referee report.

3 Course schedule

This table is the schedule as of the start of the semester. Classes are occasionally cancelled or delayed, in which case topics shift to later meetings. The course website carries the current schedule and is the version to work from. Class numbers stay attached to their topic and slides, so a shift changes which date a class falls on, not its number.

Week	Day	Class Date	Topic
Week 1	M	24-Aug-2026	Class 1: Overview
	W	26-Aug-2026	Class 2: Introduction to environmental economics
Week 2	M	31-Aug-2026	Class 3: Theory of externalities
	W	02-Sep-2026	Class 4: Theory of environmental policy
Week 3	M	07-Sep-2026	Labor Day – no class
	W	09-Sep-2026	Class 5: Instrument choice I
Week 4	M	14-Sep-2026	Class 6: Instrument choice II
	W	16-Sep-2026	Class 7: Instrument choice III
Week 5	M	21-Sep-2026	Class 8: Instrument choice IV
	W	23-Sep-2026	Class 9: Welfare analysis
Week 6	M	28-Sep-2026	Class 10: Valuation I – revealed preference
	W	30-Sep-2026	Class 11: Valuation II – property value models
Week 7	M	05-Oct-2026	Fall recess – no class
	W	07-Oct-2026	Class 12: Attend SWEEEP in lieu of class
Week 8	M	12-Oct-2026	Class 13: Guest lecture – hedonics and VSL
	W	14-Oct-2026	Class 14: Marginal cost of public funds
Week 9	M	19-Oct-2026	Class 15: Midterm (in class)
	W	21-Oct-2026	Class 16: TBD
Week 10	M	26-Oct-2026	Class 17: Topics: Utility pricing
	W	28-Oct-2026	Class 18: Topics: Water
Week 11	M	02-Nov-2026	Class 19: Topics: Electricity externalities and pricing
	W	04-Nov-2026	Class 20: Topics: Energy and smart technologies
Week 12	M	09-Nov-2026	Class 21: Topics: Climate change
	W	11-Nov-2026	Class 22: Topics: Conservation
Week 13	M	16-Nov-2026	Class 23: Topics: Forecasts
	W	18-Nov-2026	Class 24: Topics: Domestic climate policy
Week 14	M	23-Nov-2026	Class 25: Proposal presentations
	W	25-Nov-2026	Thanksgiving break – no class
Week 15	M	30-Nov-2026	Class 26: Proposal presentations
	W	02-Dec-2026	Class 27: Proposal presentations
Week 16	M	07-Dec-2026	Class 28: Proposal presentations

4 Readings

The reading list is posted on the course website at <https://caseyjwichman.com/econ7102/>, organized by week, with PDFs. The list is tentative and will change with the pacing of the course. The website is always the current version.

The primary required text is Phaneuf & Requate (2017), *A Course in Environmental Economics: Theory, Policy, and Practice*, Cambridge University Press. All other readings are provided electronically. There are no other required purchases.

Readings marked **[response due]** on the website require a written response. Readings marked **[discussion]** are eligible for a student-led discussion.